

NIKHIL SAI TATA

Software Developer | Full-Stack (React, Node.js, MySQL) | Backend & Cloud (AWS)

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SUMMARY

Software Developer and 2026 Computer Science graduate who designs, builds and deploys full-stack web applications end to end. Built and launched **KnowYC**, a live platform with **149 REST APIs**, a **29-table MySQL schema** and **role-based access control**. Skilled in JavaScript, React, Node.js/Express, Python, SQL and AWS, with a strong focus on secure, scalable and maintainable code.

TECHNICAL SKILLS

Languages: JavaScript (ES2023), Python, SQL, C++, HTML5, CSS3

Frontend: React.js, React Router, Vite, Responsive Design, Accessibility, Leaflet

Backend: Node.js, Express.js, Django, RESTful APIs, Authentication & Authorization (OAuth 2.0 / OpenID Connect, Sessions, RBAC), Zod

Databases: MySQL, Amazon RDS, Relational Database Design, Normalization, Indexing, Schema Migrations

Cloud & DevOps: AWS (Lambda, S3, ECS, RDS, IAM, CloudWatch, VPC, Bedrock), Docker, Terraform, Jenkins, Render, Git, GitHub, Linux

Tools & Practices: Postman, API Testing, Unit & Integration Testing, OWASP Top 10

AI / LLM: RAG, LangChain, LLM APIs (OpenAI, Anthropic Claude), Prompt Engineering, PyTorch, Scikit-learn, Pandas

Core CS: Data Structures and Algorithms, Operating Systems, Database Management Systems, Computer Networks

PROJECTS

KnowYC - Verified Student Community Platform | React, Node.js, Express, MySQL, REST API

Live: knowyc.org

- Built and deployed a **full-stack web platform** (React 19, Node.js/Express 5, MySQL 8) for verified college students with housing, marketplace, study resources, Q&A forum and map-based search, exposing **149 REST APIs** over a normalized **29-table MySQL schema**.
- Engineered **secure authentication and authorization**: bcrypt hashing, revocable HttpOnly-cookie sessions, email OTP, Google Sign-In (OpenID Connect), role-based access control, per-college data isolation, rate limiting and CSP headers.
- Implemented **geospatial search** with MySQL ST_Distance_Sphere and Leaflet (100 m-5 km radius), a **13,000+ college directory** search in ~30 ms, and a secure upload pipeline with PDF/DOCX text extraction and admin moderation.
- Deployed to production at **knowyc.org** on Render with a TLS-secured Aiven MySQL database and Brevo transactional email at zero hosting cost, with automated API tests covering core user flows.

Automated Document Forgery Detection using Computer Vision & ML | Python, Django, Scikit-learn

GitHub

- Developed a **full-stack document forgery detection platform** using Django, Computer Vision and Machine Learning that automatically verifies the authenticity of uploaded PDF and image documents.
- Built a feature-extraction and classification pipeline using **Error Level Analysis (ELA)**, image preprocessing and forensic analysis, achieving **95% detection accuracy** with Random Forest and 92% with SVM.
- Implemented backend routes for **document processing, metadata inspection, model inference and report generation**.
- Designed an **interactive web interface** that highlights suspected tampered regions and forensic insights, speeding up document review.

EXPERIENCE

Coding and Programming Intern | Samsung Innovation Campus (NSIC)

Sep 2024 - Oct 2024

Project: AskSQL - Natural Language to SQL Service

- Built **AskSQL**, a natural-language-to-SQL service on AWS that turns plain-English questions into SQL, runs them and returns live results, letting non-technical users query business data without manual reports.
- Designed a **serverless backend** with AWS Lambda and Amazon RDS that automates the full request lifecycle: prompt intake, SQL generation, query execution and real-time response delivery.
- Built a **schema-aware retrieval layer** with Amazon Titan Text Embeddings v2 and Bedrock Knowledge Base, improving schema understanding and reducing query ambiguity across complex relational datasets.
- Achieved **95%+ query accuracy** across 100+ advanced aggregation queries through prompt engineering and few-shot learning, making structured data accessible to non-technical stakeholders.

LEADERSHIP & CERTIFICATIONS

Google Student Ambassador

Aug 2025 - Dec 2025

- Organized and delivered **4 hands-on Generative AI workshops for 500+ students**, driving campus adoption of Google Gemini through training in prompt engineering and practical AI tools.

Postman API Fundamentals Student Expert | Postman | [Verify Credential](#)

Oct 2025

EDUCATION

Bharath Institute of Higher Education and Research, Chennai, India

2022 - 2026

Bachelor of Technology in Computer Science

CGPA: 8.44 / 10

Narayana Junior College, Hyderabad | Intermediate (Class XII), MPC | 95.6%

2020 - 2022

Ideal High School, Suryapet | SSC (Class X) | CGPA: 10 / 10

2019 - 2020